Factory Biodiversity Library: Urban Fauna Coexistence

De Almudena Urbieta

Factory Biodiversity Library: Urban Fauna Coexistence. Collection: Industrial Resilience Series. The integration of aerospace infrastructure with local ecosystems requires a paradigm shift from containment to coexistence. This technical manual, authored by Almudena Urbieta, outlines the strategic protocols for managing urban fauna within high-precision manufacturing environments, ensuring both operational safety and environmental compliance through scientific mitigation.

The strategic context is rooted in the mitigation of operational and legal risks. Uncontrolled fauna presence can lead to Foreign Object Damage (FOD) incidents and unscheduled downtime in the supply chain. Elias analyzes risk maps where avian migratory routes intersect with engine testing zones. Compliance with current environmental regulations demands a technical management approach that prevents sanctions and optimizes infrastructure resilience against biological vectors.

The project's scientific and technical framework is supported by applied biology and material physics. Inés, an ecosystem services specialist, defines the parameters of native biodiversity that function as natural regulators. Species behavior is studied to predict nesting and transit patterns. The goal is not eradication, but the channeling of fauna toward low-interference zones through the design of controlled habitat gradients.

Field interference analysis identifies birds as the primary critical factor in daily operations. Nests within HVAC systems and hangar structures pose risks of obstruction and corrosion from biological waste. Elias supervises the integrity of air intakes and external sensors. Mitigation engineering requires the identification of specific acoustic frequencies and visual stimuli that displace these populations without compromising the plant's communication systems.

Interferences from terrestrial fauna primarily affect underground infrastructure and cabling systems. Inés documents incursions by micro-mammals and reptiles seeking shelter in electrical conduits. Damage caused by the mastication of polymers can lead to short circuits in critical systems. Technical seals and physical barriers made of non-degradable materials are established to prevent access to the plant's data and power nodes.

Solution engineering proposes the use of non-lethal deterrent systems based on ultrasonic and laser technology. Elias verifies the installation of automated emitters in taxiway areas. These maintenance protocols ensure that devices operate within frequency ranges permitted by aeronautical regulations, preventing electromagnetic interference with the navigation systems of aircraft under testing.

Habitat management is the long-term solution to reduce the plant's attractiveness to opportunistic fauna. Inés directs the replacement of ornamental flora with low-profile native biodiversity that does not produce fruits or seeds attractive to birds. This redesign of surrounding ecosystem services creates a natural "biological exclusion zone," minimizing the need for constant mechanical interventions.

It is imperative to distinguish industrial reality from common superficial marketing myths.